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CO1-ASSESMENT TOOL 1-APPLICATION BASED QUESTIONS

DSA0308-NLP

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1.Resume Information extraction using regular expression.

Explanation:

1. We use python re model to extract the information from the resume.
2. Then, we have to import the python re model.
3. We define the regular expression pattern to identify the candidate details such as name, email address, mobile number, technical skills, and years of experience.
4. Then, we use the regular patterns to search the resume and extract the information which is required.
5. We identify the technical skills such as Python, Java, SQL, Machine Learning, and NLP.
6. Now, we extract the candidate's years of experience by the numbers which are next to the words "year" or "years".
7. Then we check whether the candidate has experience of two years and Python skills.
8. If the candidate has both the skills and the 2 years of experience he will be eligible, otherwise he is not eligible.

2.Product searching system using regular expressions.

Explanation:

1. We use Python's re module to search for products using regular expressions.
2. First, we create a list of the available products.
3. Then, we accept the search keyword and the type of search the user wants to perform.
4. For exact keyword searches we match the exact product name to the keyword.
5. To prefix searches, we find the products which names begin with the given keyword.
6. To the suffix searches, we find the products which names ends with the given keyword.
7. To the partial keyword searches, we find the products which contains the given keyword.
8. we use the case insensitive matching method to find the products which are in uppercase and lowercase.
9. Finally, we display all the matching product names and calculate the total number of matching names.

3. University Registration System Using Regular Expressions.

Explanation:

1. We use the python re model to check the student registration details.
2. We collect the students register number, institutional email address, course code, semester, and mobile number.
3. Then, we define the regular expression patterns according to the required format.
4. We define regular expression patterns according to the required format of each field.
5. Then, we check the register number whether it follows the required format or not.
6. We check whether the institutional email address contains the correct domain.
7. We validate the course code using a pattern which matches the required structure.
8. We validate the mobile number to see that it follows the required format or not.
9. Then, the system will display a separate validation message for each field, indicating whether the entered information is correct or in correct.
10. If all fields are valid, the system displays a successful registration message. Otherwise, it displays an unsuccessful message.